

Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_best_single_fixed_heuristic.
- Best combined transfer gap: phase8_oracle_selector.
- Lowest selector regret: phase8_best_single_fixed_heuristic.
- Pareto-efficient conditions: phase8_random_portfolio, phase8_supervised_ml_selector.

Run Metadata

- run_name: run_20260520_213122_I
- started_at_local: 2026-05-20 21:31:22
- finished_at_local: 2026-05-20 21:46:24
- duration_hhmm: 00:15
- duration_seconds: 902.662
- seed_offset: 11000
- replicate_label: I
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	700.194257	0.07226	0.007284	0.031222	0.0	0.0	False
phase8_random_portfolio	random_portfolio	none	0.169899	0.097639	78.5905	0.169899	0.078717	0.11231	0.0	0.0	True
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	700.106143	0.07226	8.9e-05	0.026678	0.0	0.0	False

Condition	Mode	Replay	Final TSPLI B Gap	Selector Regret	Runtime (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	696.790643	0.07226	0.007284	0.03122	0.0	0.0	True
phase8_llm_static_selector	static_selector	none	0.086887	0.014627	1347.321329	0.167189	0.002791	0.033774	0.0	0.62	False
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.190914	0.118654	104.241157	0.190914	0.036509	0.093395	0.536232	0.62	False
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.124922	0.052662	1852.313071	0.330472	0.014601	0.055245	0.587454	0.6975	False
phase8_full_solver_evolution	full_solver	none	0.213387	0.141127	100.055386	0.213387	0.150269	0.173523	0.852585	0.56	False

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 700.194257 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.169899, selector regret 0.097639, runtime-adjusted gap 0.169899.
- Family transfer 0.078717 and combined transfer 0.11231.
- Runtime 78.5905 ms, runtime inflation -0.887759, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 700.106143 ms, runtime inflation -0.000126, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 696.790643 ms, runtime inflation -0.004861, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.086887, selector regret 0.014627, runtime-adjusted gap 0.167189.
- Family transfer 0.002791 and combined transfer 0.033774.
- Runtime 1347.321329 ms, runtime inflation 0.924211, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: annealed_multi_start:cluster_first_local_search:farthest_insertion_two_opt:annealed_multi_start:stag3:cand3:restart1.
- Controller rule summary: {"acceptance_bias": 0.005, "candidate_limit_offset": 3, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "edge_preserving_restart", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive selector over a frozen TSP heuristic portfolio using structure descriptors; tuned to favor farthest/compact 2-opt on general cases, candidate-pruning for grid-like, and clustered/bottleneck-aware search only when justified by de", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.08, "name": "det_heur_portfolio_instance_adaptive_v1", "nearest_neighbor_trap_heuristic": "farthest_insertion_two_opt", "random_like_heuristic": "annealed_multi_start", "restart_offset": 1, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 3, "temperature_scale": 1.1, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.190914, selector regret 0.118654, runtime-adjusted gap 0.190914.
- Family transfer 0.036509 and combined transfer 0.093395.
- Runtime 104.241157 ms, runtime inflation -0.851125, mean novelty 0.536232, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:edge_preserving_restart:candidate_pruned_two_opt:stag4:cand-1:restart1.
- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": -1, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Interpretable deterministic instance-adaptive hyper-heuristic controller over a frozen TSP heuristic portfolio using instance descriptors and online stagnation/progress signals to schedule bounded tuning.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt",

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"low_improvement_threshold": 0.1, "name": "deterministic_instance_adaptive_tsp_hh_epoch2",
"nearest_neighbor_trap_heuristic": "annealed_multi_start", "random_like_heuristic": "nearest_neighbor_multistart",
"restart_offset": 1, "stagnation_switch_heuristic": "candidate_pruned_two_opt", "stagnation_threshold": 4,
"temperature_scale": 1.25, "time_budget_trigger": 0.6, "two_cluster_bottleneck_heuristic":
"edge_preserving_restart"}

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phase8_ilm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.124922, selector regret 0.052662, runtime-adjusted gap 0.330472.
- Family transfer 0.014601 and combined transfer 0.055245.
- Runtime 1852.313071 ms, runtime inflation 1.645427, mean novelty 0.587454, and complexity 0.6975.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:edge_preserving_restart:cheapest_insertion_two_opt:stag3:cand1:restart0.
- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": 1, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive selector over frozen heuristic portfolio with bounded online scheduling. Structure-driven initial heuristic choice (grid/cluster/bottleneck/corridor/trap) and stagnation/failed-perturbation driven deterministic switching to a si", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.14, "name": "det_adaptive_tsp_portfolio_v3", "nearest_neighbor_trap_heuristic": "candidate_pruned_two_opt", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 0, "stagnation_switch_heuristic": "cheapest_insertion_two_opt", "stagnation_threshold": 3, "temperature_scale": 0.95, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.213387, selector regret 0.141127, runtime-adjusted gap 0.213387.
- Family transfer 0.150269 and combined transfer 0.173523.
- Runtime 100.055386 ms, runtime inflation -0.857103, mean novelty 0.852585, and complexity 0.56.
- Pareto efficient: False.

Judge Appendix

Conservative Interpretation of TSP Adaptive Heuristic Portfolio Suite Results

Criterion	Best Condition(s)	Notes & Interpretation
Held-out TSPLIB Gap (Primary)	phase8_best_single_fixed_heuristic, phase8_supervised_ml_selector, phase8_oracle_selector	All three show identical final TSPLIB gaps: 0.07226. The fixed heuristic (farthest_insertion_two_opt) is as good as the ML selector, matching the oracle upper bound closely.
Selector Regret vs Oracle	phase8_best_single_fixed_heuristic, phase8_supervised_ml_selector (0.0 regret)	Adaptive selectors achieved *no regret* compared to oracle selector, indicating interpretable and effective instance-adaptive control.

Criterion	Best Condition(s)	Notes & Interpretation
Runtime-Adjusted Gap	phase8_best_single_fixed_heuristic, phase8_supervised_ml_selector, phase8_oracle_selector (≈ 0.072)	Despite runtime inflation differences, gaps are close, with the fixed and supervised selectors achieving similar overall efficiency.
Runtime Inflation	The fixed heuristic and supervised selector have zero or negative runtime inflation; static and adaptive (LLM evolved) controllers show up to near 1.6x inflation (e.g., phase8_llm_static_selector, phase8_llm_evolved_controller_diversity_failure_replay)	Runtime inflation is a concern for more complex adaptive controllers; simpler ML selector maintains runtime efficiency without gap loss.
Pareto Efficiency	phase8_random_portfolio and phase8_supervised_ml_selector	ML selector is pareto-efficient combining good gap with acceptable runtime, unlike fixed heuristic which is near but not marked pareto efficient. Random portfolio is not competitive in gap.
Cross-Family Transfer Gap	Lowest transfer gap with phase8_oracle_selector (0.0267), close second phase8_best_single_fixed_heuristic and phase8_supervised_ml_selector (~ 0.0312)	ML selectors and best single fixed heuristic generalize well to unseen families, indicating robust instance-adaptive control. Adaptive controllers show larger transfer gaps.
Adaptation Efficiency & Complexity	Adaptive controllers have modest adaptation efficiency (~ 0.06 - 0.09), but runtime inflation and increased complexity (~ 0.6 - 0.7) without meaningful TSPLIB gap improvements	Adaptive control is interpretable but not clearly beneficial over simpler ML approach with respect to primary metric, and adds complexity and runtime costs.
Controller Novelty	Adaptive controllers show non-zero novelty (> 0.5), but this does not translate to improved held-out TSPLIB gap or regret	Suggests that exploration/new code does not improve final portfolio control effectiveness in this phase.

Summary

- ****Primary endpoint (held-out TSPLIB gap):**** The best single fixed heuristic (farthest_insertion_two_opt) and the supervised ML selector both achieve the lowest gap (0.07226), matching the oracle selector upper bound. This implies no advantage yet from adaptive hyper-heuristic control in terms of solution quality.
- ****Instance-adaptive control:**** ML selector offers interpretable instance-adaptive control without runtime inflation or selector regret, showing strong practical promise. Adaptive LLM-evolved controllers, though interpretable, do not improve on this baseline and incur large runtime and complexity costs.
- ****Runtime and efficiency:**** Prefer simpler, static or ML selectors for deployment due to near-oracle gap performance and controlled runtime overhead; avoid adaptive controllers causing runtime inflation with no gap benefit.
- ****Cross-family transfer:**** The ML selector and fixed baseline transfer well to unseen problem families, indicating robustness and generalizability of selected heuristics.
- ****No new algorithm discovery:**** The results focus on controlling known heuristics; no evidence of new heuristics or fundamentally novel algorithms is observed.

Recommendations

- Prioritize ML-based supervised selector for deployment as an interpretable, low-regret, efficient instance-adaptive heuristic controller.
- Avoid or further refine adaptive LLM-based controllers until they can demonstrate gap improvements or runtime efficiency closer to oracle/fixed baseline.
- Continue to monitor transfer gaps to ensure robustness across diverse instance families without runtime penalty.
- Focus future research on improving adaptation efficiency and runtime rather than code novelty or complexity increase.